The MABEL Motion Library: authoring whole-robot clips against a rest pose that is not the origin

MABEL project

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Abstract

The choreography timeline composes long-horizon robot behaviour from a library of clips: expressive animations, scripted task policies, parametric navigation primitives and static poses. Every clip is a closure mapping clip-local time to an absolute whole-body configuration, and every clip was authored with its envelopes vanishing at the clip edges — so that dropping any clip anywhere is safe. That construction silently assumes the frame the envelopes vanish *into* is the robot's rest pose. It is not: MABEL's zero configuration is a 91.3° bent elbow costing 38.6 N m to hold, and the true rest pose sits 1.52 rad away at the elbow. Playing the library raw against a correct rest pose would command that difference on a clip's first 50 ms tick: 30 rad s^{-1}, twelve times the library's own budget. We introduce a *rest anchor* — a single eased interpolation, applied at the one point where a clip closure is evaluated, that carries the pose out of rest and back into it — whose ramp is derived from the clip's own travel so that the anchor cannot break the velocity budget it is protecting. The library is then expanded from 44 to 81 entries (32 animations, 20 static poses, 18 task policies, 10 navigation primitives, one transition) and audited against the robot model: every one of the 60 sampled clips starts and ends within 0 rad of rest, none leaves a joint limit, the worst commanded velocity is 2.43 rad s^{-1} against a 2.5 rad s^{-1} budget, and none produces a self-collision except the one primitive whose target the model cannot represent.

Keywords: motion library, keyframe animation, rest pose, velocity limiting, trajectory blending, robot choreography, teleoperation.

1 Overview

The library lives in `web/js/motions.js` and is the single source of truth for what MABEL can be asked to do on a timeline. Two generators consume it without reimplementing it: `gen_action_catalog.py` ports the metadata to Swift, and `gen_motion_clips.py` runs the registry under node and samples every clip at the executor's own 20 Hz, producing the table the iPhone and Vision Pro applications play. Because the studio's command stream is itself 20 Hz and the servo ramps between commands, that table is not an approximation of the studio's motion — at the rate that reaches hardware it *is* the motion.

This document specifies how a clip is authored (§2), derives the rest anchor and its ramp (§3), reports the audit of the whole library against the model (§4), and records what the audit found (§6). The rest pose itself is derived in `controller/docs/rest_pose.tex`; the guard that protects the junction *between* two blocks is `docs/timeline_safety.tex`. Neither is repeated here.

2 Method: what a clip is

A clip is a record carrying metadata (title, lane, category, duration, whether it may be held) and up to two closures,

$$J : [0, d] \to \mathbb{R}^n, \qquad N : [0, d] \to [-1, 1]^4, \tag{1}$$

the first mapping clip-local time to an absolute joint configuration and the second to a navigation joystick stream. Clips are written from a small easing toolkit — a raised-cosine ramp σ, a windowed envelope w that rises, holds and falls, a windowed oscillation, and a breathing term — so that every authored term is a product of an amplitude and an envelope that vanishes at both edges. A clip that raises an arm therefore returns the arm by construction, and the resulting motion is C^1 at both ends.

2.1 The frame problem

The consequence the authors relied on is $J(0) = J(d) = \mathbf{0}$. That is a statement about the *origin of the joint space*, not about the robot, and it is only useful if the origin is where the robot rests. On MABEL it is not (`rest_pose.tex`): $q = 0$ folds both elbows to a right angle. Every authored silhouette was validated in that frame and is correct in it; what is wrong is only the frame's endpoints.

Re-authoring thirty-seven elbow and shoulder-roll terms across the library would mean re-validating every silhouette against the simulator, and would leave the next author with the same trap. Instead we change the frame once.

3 The rest anchor

Let $r \in \mathbb{R}^n$ be the rest pose. Define the clip's commanded pose as

$$\hat{J}(t) \;=\; r \;+\; \big(J(t) - r\big)\, a(t), \tag{2}$$

with the anchor envelope

$$a(t) \;=\; \begin{cases} \sigma(t/\rho)\,, & \text{clip is } \texttt{hold}, \\ w(t;\, 0,\, d,\, \rho), & \text{otherwise}, \end{cases} \qquad \sigma(u) = \tfrac{1}{2} - \tfrac{1}{2}\cos(\pi\, \mathrm{clip}_{[0,1]}(u)). \tag{3}$$

Two properties make (2) the right construction rather than merely a workable one. At the edges $a = 0$, so $\hat{J} = r$ *exactly, whatever the clip returned there* — the guarantee no longer depends on the author getting the envelopes right. Across the body $a = 1$, so $\hat{J} = J$ exactly, and every silhouette validated in the old frame is reproduced unchanged. A `hold` clip ramps a up and keeps it, because holding the final pose is precisely what that flag means.

3.1 Sizing the ramp

The anchor is a blind interpolation across a distance the author did not choose, so its duration cannot be a taste parameter. Let

$$\Delta \;=\; \max_{t \in [0,d]} \; \max_{k} \; |J_k(t) - r_k| \cdot s_k \tag{4}$$

be the farthest the anchor must travel, where s_k converts the two non-radian axes (a grip unit, a metre of lift) into radian-equivalents. A raised cosine reaches $\kappa = \pi/2$ times its average rate, so holding the anchor to a ceiling v_a requires

$$\rho \;=\; \max\!\left(\rho_{\min},\; \min\!\left(\rho_{\max},\; \frac{\kappa\Delta}{v_a}\right)\right), \qquad \rho_{\max} = \begin{cases} d & \texttt{hold} \\ d/2 & \text{otherwise,} \end{cases} \tag{5}$$

with $v_a = 1\,\mathrm{rad\,s^{-1}}$ — the same ceiling the pose-jump guard uses, because this is the same kind of object: an automatic transition across a distance nobody authored. Δ is sampled once per clip and memoised.

3.2 Static poses become one number

A pose clip's J is constant, so (5) can be evaluated before the clip exists and its *duration* derived rather than guessed: $d = \kappa\Delta/v_a + \tau_{\text{tail}}$ with a 1.2 s tail. A pose is then nothing but its target; the anchor supplies the transition. `pose_tuck` is a 1.5 s block and `pose_arms_up` a 6.0 s one, which is simply how far each has to travel at the ceiling.

3.3 What is exempt

The `blend` transition is not anchored. It is not authored in the joint origin; it interpolates between two *measured* poses handed to it by the pose-jump guard, and anchoring it to rest would erase exactly the transition it exists to make.

4 Experiments

All numbers are produced by `tools/exp_pose_library.py`, which drives the real registry through node and does the kinematics in MuJoCo against `simulation/mabel_mujoco/models/mabel_full.xml`. Nothing is hand-entered.

4.1 E1 — the anchor holds exactly

Across the 60 sampled clips the worst first-sample distance from rest is 0 rad and the worst last-sample distance for a non-`hold` clip is 0 rad: the guarantee of (2) is algebraic, not approximate, and the audit confirms no clip evades it. Twenty-three clips end away from rest, and all twenty-three are `hold` clips whose whole purpose is to strand the robot somewhere — the twenty poses, the two lift primitives, `pickup` (which finishes carrying what it picked up) and `sleep` (which stays asleep).

4.2 E2 — the budget survives the anchor

Sampled at the executor's 20 Hz, the worst commanded joint speed in the library is $2.43\,\mathrm{rad\,s^{-1}}$ (`celebrate`, the lift) against a $2.5\,\mathrm{rad\,s^{-1}}$ budget, with a mean across clips of $1.45\,\mathrm{rad\,s^{-1}}$. Anchor ramps run from 0.5 s (`pose_home`, which has nowhere to go) to 4.71 s (`pose_arms_up`, which travels 3.0 rad).

4.3 E3 — pose silhouettes

Each of the twenty pose targets was chosen against the measured silhouette rather than by eye. Table 1 gives a representative subset: the hand's displacement from its shoulder, the residual elbow bend, and the static holding torque the pose costs.

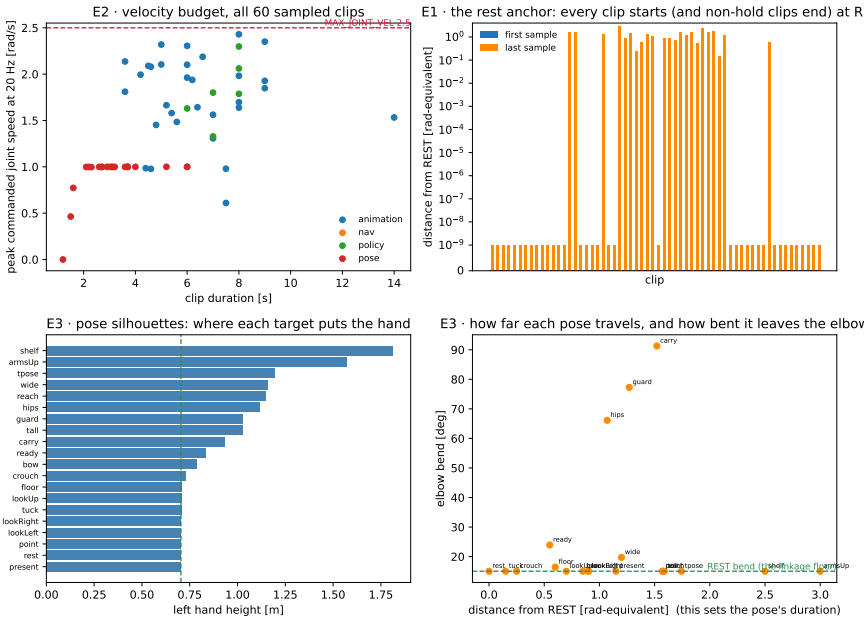


Figure 1: **E2** (top left) peak commanded joint speed against clip duration for all 60 sampled clips, by category, against the library budget. **E1** (top right) the rest anchor: the first and last sample of every clip, as a distance from the rest pose; non-hold clips are zero at both ends. **E3** (bottom) the twenty static pose targets — where each puts the hand, and how its travel from rest sets both its block duration and its residual elbow bend.

Table 1: Selected pose targets, measured. Displacements are the left hand relative to its shoulder in the robot's own frame. *Bend* is the upper-arm/forearm angle; 15.1° is the linkage's floor.

Pose	fwd [m]	out [m]	up [m]	bend [°]	hand z [m]	$\sum \lvert\tau\rvert$ [N m]
`rest`	0.013	0.019	−0.369	15.1	0.704	1.64
`tall`	0.068	0.019	−0.363	15.1	1.027	9.41
`tpose`	0.014	0.367	0.045	15.1	1.196	38.95
`reach`	0.369	0.023	0.010	15.1	1.150	58.56
`armsUp`	0.009	0.040	0.368	15.1	1.573	3.85
`shelf`	0.210	0.018	0.305	15.1	1.815	31.79
`carry`	0.216	0.044	−0.146	91.3	0.933	46.30
`hips`	0.133	0.285	0.015	66.1	1.118	56.01
`bow`	−0.269	0.019	−0.254	15.1	0.787	38.18

Three facts from the probe drove the re-derivation of the older targets. The shoulder pitch sweeps the arm in the sagittal plane with the elbow extended (left 0 hanging, -1.571 straight forward, -3.0 overhead); the shoulder roll abducts, with $+1.571$ putting the hand 0.367 m outboard; and the neck is drivable from a joint command, which the library had previously been told it was not. The old `armsUp` target of -1.75 had been chosen when the elbow rested folded, so it pointed the *forearm* up while the upper arm stayed forward — with a straight elbow the same number points the whole arm forward, which is why every pose target was re-measured rather than carried over.

4.4 E4 — limits and contacts

No clip in the library leaves a joint's MJCF limit at any sample. One clip reaches a self-contact: `lift_down`, which asks the column for zero (§6). Fixing the others required re-expressing six clips whose lift terms had been authored when "lift 0" meant "down" — under a rest lift of 0.158 75 m those absolute values are *below* rest, and below the model's collision floor — and widening two clips whose hands met in front of the face or brushed the lift column.

5 Implementation and usage

```
node --test web_gui/control_center/tests/js/motions.test.mjs   # the gates
python3 web_gui/control_center/tools/exp_pose_library.py       # this document
python3 simulation/tools/gen_action_catalog.py                 # -> both apps
python3 simulation/tools/gen_motion_clips.py                   # -> both apps
python3 ios_apps/iOS_v2/tools/exp_timeline_motion.py           # port fidelity
```

Adding a clip means adding an entry to `MOTION_CLIPS`; adding a pose means adding a target to `POSE_TARGETS` and one line to the `poses(...)` table. Both generators must then be re-run — the applications carry no motion math of their own beyond the parametric navigation primitives and the blend, and `--check` on the baker fails the build if the committed table has drifted.

Four properties are asserted by `tests/js/motions.test.mjs` on every clip: it stays inside the model's joint ranges; it stays under `MAX_JOINT_VEL` at 20 Hz with the grip and lift axes converted to radian-equivalents; it starts at rest and, unless it holds, ends there; and it never commands the lift below the model's usable floor.

6 Limitations

The retracted lift is not representable. The model's lift column self-intersects the chassis below 0.155 m per stage (116 contacts when fully retracted), so only the upper half of the declared stroke is usable in simulation. `lift_down` still asks for zero, because that is a real capability of the real column and clamping it here would quietly remove half the stroke from the hardware to work around a collision-geometry error. The test that forbids every other clip from going there exempts it by name.

The anchor can be squeezed. (5) caps ρ at the block's own duration. A clip authored with a short duration and a distant target will therefore exceed the $1\,\mathrm{rad\,s^{-1}}$ anchor ceiling, up to the $2.5\,\mathrm{rad\,s^{-1}}$ clip budget. The static poses derive their duration so this cannot happen to them; an animation author must still check E2.

Silhouettes are geometric, not aesthetic. The probe reports where the hand is, the elbow angle, the holding torque, limit violations and contacts. It cannot report whether `shrug` reads as a shrug. Every animation in the library was measured; only the nine that predate this work have also been watched on the simulator.

Self-collision is checked at samples, not swept. Contacts are evaluated on a subsampled grid over each clip. A brief interpenetration between two grid points would be missed. No continuous-collision check is performed, and the hands' finger joints are the geometry most likely to hide one.

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